



ECEN 314 - Signals and Systems

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Module 8 – LTI Systems with Sinusoidal Inputs

Learning Objectives



- Be able to find the output of any LTI system if the input is sinusoidal.
- Understand the difference between a phasor transform and a Laplace transform.

LTI Systems with Sinusoidal Inputs



In previous slides, it was determined that a linear, time-invariant (LTI) system can be characterized by:

- ✓ Impulse Response – In which case the input/output relationship in the time domain is written as a convolution integral

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(v)h(t - v)dv$$

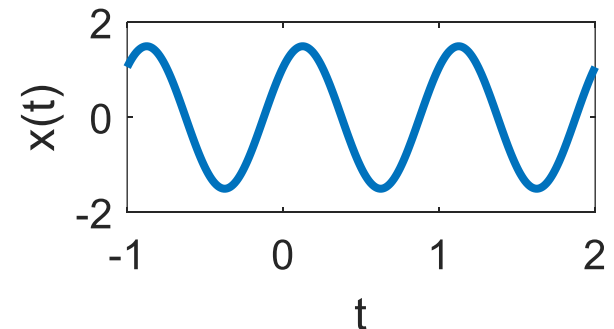
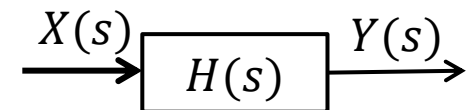
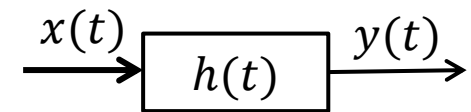
- ✓ Transfer Function – In which case the input/output relationship in the s-domain is written as a multiplication

$$Y(s) = X(s)H(s)$$

In this set of slides, we will explore what happens to our LTI systems when the inputs are sinusoidal:

$$x(t) = A\cos(\omega t + \theta)$$

$$X(s) = ???$$



LTI Systems with Sinusoidal Inputs



Since the sinusoids we are talking about here are two-sided, they do not have an s-domain representation. We will have to use the time domain description of our system to determine the output of the system.

To start with, consider an LTI system, described by some impulse response, $h(t)$, with an input that is a complex exponential, $x(t) = Ae^{j(\omega t + \theta)}$.

$$\begin{aligned} x(t) &\longrightarrow \boxed{h(t)} \longrightarrow y(t) = \int_{-\infty}^{\infty} h(v)x(t-v)dv \\ y(t) &= A \int_{-\infty}^{\infty} h(v)e^{j(\omega(t-v) + \theta)} dv = Ae^{j(\omega t + \theta)} \int_{-\infty}^{\infty} h(v)e^{-j\omega v} dv \end{aligned}$$

Assuming the system is causal so that $h(t)$ is right-sided ($= 0$ for $t < 0$), then

$$y(t) = Ae^{j(\omega t + \theta)} \int_0^{\infty} h(v)e^{-j\omega v} dv = Ae^{j(\omega t + \theta)} H(s)|_{s=j\omega} = A \cdot H(j\omega) \cdot e^{j(\omega t + \theta)}.$$

If we write $H(j\omega)$ in polar form, $H(j\omega) = |H(j\omega)|e^{j\angle H(j\omega)}$, then

$$y(t) = A \cdot |H(j\omega)| \cdot e^{j(\omega t + \theta + \angle H(j\omega))}.$$

The output is a complex exponential of the same frequency as the input. Only the amplitude and phase have been affected.

LTI Systems with Sinusoidal Inputs



The result of the previous slide can be generalized to real sinusoids since a sine or cosine can be constructed from complex exponentials.

$$x_1(t) = e^{j\omega t} \longrightarrow \boxed{h(t)} \longrightarrow y_1(t) = H(j\omega) \cdot e^{j\omega t}$$

$$x_2(t) = e^{-j\omega t} \longrightarrow \boxed{h(t)} \longrightarrow y_2(t) = H(-j\omega) \cdot e^{-j\omega t}$$

$$x(t) = A \cos(\omega t + \theta) = \frac{A}{2} e^{j(\omega t + \theta)} + \frac{A}{2} e^{-j(\omega t + \theta)} = \frac{A}{2} e^{j\theta} e^{j\omega t} + \frac{A}{2} e^{-j\theta} e^{-j\omega t}$$

Using the linearity property, the output corresponding to $x(t) = A \cos(\omega t + \theta)$ will be:

$$y(t) = \frac{A}{2} e^{j\theta} H(j\omega) e^{j\omega t} + \frac{A}{2} e^{-j\theta} H(-j\omega) e^{-j\omega t}$$

If the system is real, then $H(-j\omega) = H^*(j\omega)$, in which case the second term above is the complex conjugate of the first. Recall that $z + z^* = 2\text{Re}[z]$ so that

$$y(t) = A \cdot \text{Re}[H(j\omega) e^{j(\omega t + \theta)}] = A \cdot |H(j\omega)| \cdot \cos(\omega t + \theta + \angle H(j\omega)).$$

LTI Systems with Sinusoidal Inputs



The result developed in the last few slides forms the foundation for much of what we will cover for the rest of the course. We have proven the following result.

Theorem: For any real, causal, linear, time-invariant system, sinusoidal inputs will produce sinusoidal outputs of the same frequency. The amplitude will be scaled by the magnitude of the transfer function, and the phase will be shifted by the angle of the transfer function.

- This result is true for any type of sinusoid, whether it be a sine, cosine, or a complex exponential.
- For those comfortable with concepts from linear algebra:
 - ✓ Sinusoids are the eigenfunctions for LTI systems.
 - ✓ The transfer function then plays the role of the eigenvalue.
- Note that this result only applies to two-sided sinusoids, i.e., ones that persist to infinity in both directions.
- The important take-away from this result is that it is particularly easy to find the output of an LTI system when the inputs are (two-sided) sinusoids.

LTI Systems with Sinusoidal Inputs



Example 8.1: A certain system has a transfer function given by $H(s) = \frac{s+1}{s^2+5s+6}$.

Find the output of the system for each of the following inputs:

a) $x_a(t) = 2e^{jt}$.

b) $x_b(t) = \cos(3t)$.

c) $x_c(t) = 4\sin(3t + 30^\circ)$.

LTI Systems with Sinusoidal Inputs



Example 8.2: A certain system has an impulse response given by $h(t) = 3e^{-t} \sin(2t) u(t)$. Find the output of the system if the input is $x(t) = 2\sin(t - 45^\circ)$.

LTI Systems with Sinusoidal Inputs



Example 8.3: A certain system is described by the differential equation

$$y''(t) + y'(t) + y(t) = x'(t).$$

Find the output of the system if the input is $x(t) = -3\cos(2t)$.

Phasors



In electrical engineering, it is common to represent sinusoidal waveforms in terms of complex numbers known as phasors. Doing so can greatly simplify the analysis.

A general sinusoidal waveform can be written as

$$x(t) = X_o \cos(\omega_o t + \theta) = \text{Re}[X_o e^{j(\omega_o t + \theta)}] = \text{Re}[X_o e^{j\theta} e^{j\omega_o t}]$$

Let X be the complex number $X_o e^{j\theta}$. Then:

$$x(t) = \text{Re}[X e^{j\omega_o t}]$$

The diagram shows the equation $x(t) = \text{Re}[X e^{j\omega_o t}]$ with two green arrows pointing upwards. The first arrow points from the text 'sinusoidal signal' to $x(t)$. The second arrow points from the text 'Phasor' to X .

sinusoidal
signal

Phasor

Note that the magnitude and angle of the phasor represent the amplitude and phase of the sine wave.

System Analysis Using Phasors



Consider any generic system where the relationship between the input and output can be described by a differential equation.

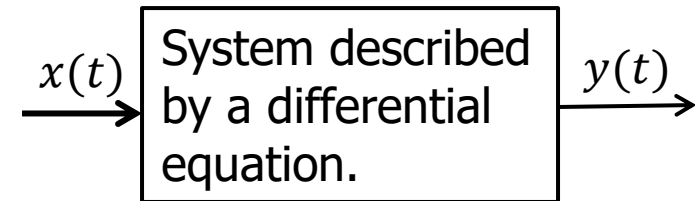
$$b_n \frac{d^n y}{dt^n} + \dots + b_2 \frac{d^2 y}{dt^2} + b_1 \frac{dy}{dt} + b_0 y(t) = a_m \frac{d^m x}{dt^m} + \dots + a_2 \frac{d^2 x}{dt^2} + a_1 \frac{dx}{dt} + a_0 x(t)$$

When the input (and hence the output) are sinusoidal, we can represent everything in terms of phasors

$$[b_n(j\omega)^n + \dots + b_2(j\omega)^2 + b_1 j\omega + b_0]Y = [a_m(j\omega)^m + \dots + a_2(j\omega)^2 + a_1 j\omega + a_0]X$$

$$H(\omega) = \frac{Y}{X} = \frac{a_m(j\omega)^m + \dots + a_2(j\omega)^2 + a_1 j\omega + a_0}{b_n(j\omega)^n + \dots + b_2(j\omega)^2 + b_1 j\omega + b_0} = \frac{a(j\omega)}{b(j\omega)}$$

Working in the phasor/frequency domain allows us to solve algebraic equations rather than differential equations.



Phasor Domain:

$$\begin{aligned} x(t) &= \text{Re}[X e^{j\omega t}], \\ \frac{dx}{dt} &= \text{Re}[j\omega X e^{j\omega t}], \\ \frac{d^2 x}{dt^2} &= \text{Re}[(j\omega)^2 X e^{j\omega t}], \\ &\vdots \\ \frac{d^m x}{dt^m} &= \text{Re}[(j\omega)^m X e^{j\omega t}], \\ y(t) &= \text{Re}[Y e^{j\omega t}], \\ \frac{dy}{dt} &= \text{Re}[j\omega Y e^{j\omega t}], \\ \frac{d^2 y}{dt^2} &= \text{Re}[(j\omega)^2 Y e^{j\omega t}], \\ &\vdots \\ \frac{d^n y}{dt^n} &= \text{Re}[(j\omega)^n Y e^{j\omega t}]. \end{aligned}$$

System Analysis Using Phasors



It should be noted that these concepts are very similar to what we did with Laplace Transforms.

$$b_n \frac{d^n y}{dt^n} + \cdots + b_2 \frac{d^2 y}{dt^2} + b_1 \frac{dy}{dt} + b_0 y(t) = a_m \frac{d^m x}{dt^m} + \cdots + a_2 \frac{d^2 x}{dt^2} + a_1 \frac{dx}{dt} + a_0 x(t)$$

Phasor
Transform

$$H(\omega) = \frac{Y}{X} \\ = \frac{a_m(j\omega)^m + \cdots + a_2(j\omega)^2 + a_1 j\omega + a_0}{b_n(j\omega)^n + \cdots + b_2(j\omega)^2 + b_1 j\omega + b_0}$$

"Frequency" Domain

$s = j\omega$

Laplace
Transform

$$H(s) = \frac{Y(s)}{X(s)} \\ = \frac{a_m s^m + \cdots + a_2 s^2 + a_1 s + a_0}{b_n s^n + \cdots + b_2 s^2 + b_1 s + b_0}$$

Laplace Domain

Note: Since the sinusoids we are now dealing with are not one-sided, we cannot really handle them with a Laplace Transform.

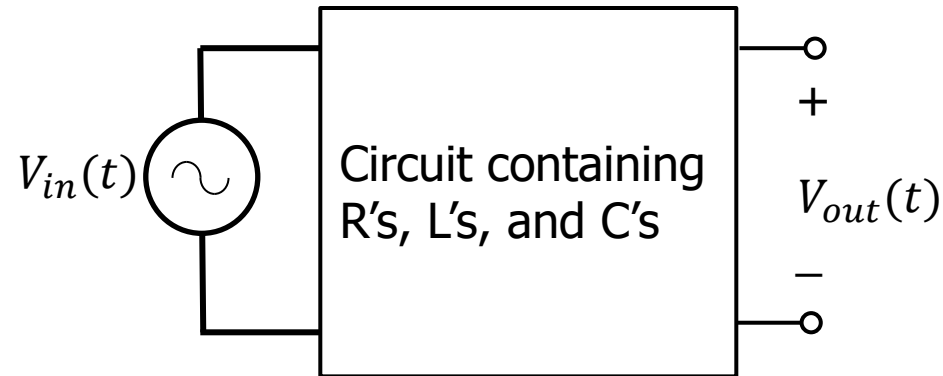
Application: Circuits



- If a sinusoidal (AC) input $V_{in}(t)$ is applied to a circuit and we measure an output voltage (or current), $V_{out}(t)$, somewhere in the circuit, the output will be a sinusoid of the same frequency, but not necessarily the same amplitude and/or phase.

$$V_{in}(t) = V_i \cos(\omega t + \theta_i),$$
$$V_{out}(t) = V_o \cos(\omega t + \theta_o).$$

- The ratio of the output voltage to the input voltage, V_o/V_i will depend on the frequency of the input and is referred to as the *magnitude response* of the circuit.
- The difference between the phase at the output and the input, $\theta_o - \theta_i$, will also depend on frequency and is called the *phase response* of the circuit.



In the phasor domain:

- $V_{in} = V_i e^{j\theta_i}$, $V_{out} = V_o e^{j\theta_o}$
- Transfer function:

$$H(\omega) = \frac{V_{out}}{V_{in}}$$

- Magnitude response:

$$|H(\omega)| = \left| \frac{V_{out}}{V_{in}} \right| = \frac{|V_o e^{j\theta_o}|}{|V_i e^{j\theta_i}|} = \frac{V_o}{V_i}$$

- Phase response:

$$\angle H(\omega) = \angle \left(\frac{V_{out}}{V_{in}} \right)$$
$$= \angle V_o e^{j\theta_o} - \angle V_i e^{j\theta_i} = \theta_o - \theta_i$$

Application: Circuits



Example 8.4: For the series RLC circuit shown, if the output voltage is measured on the capacitor,

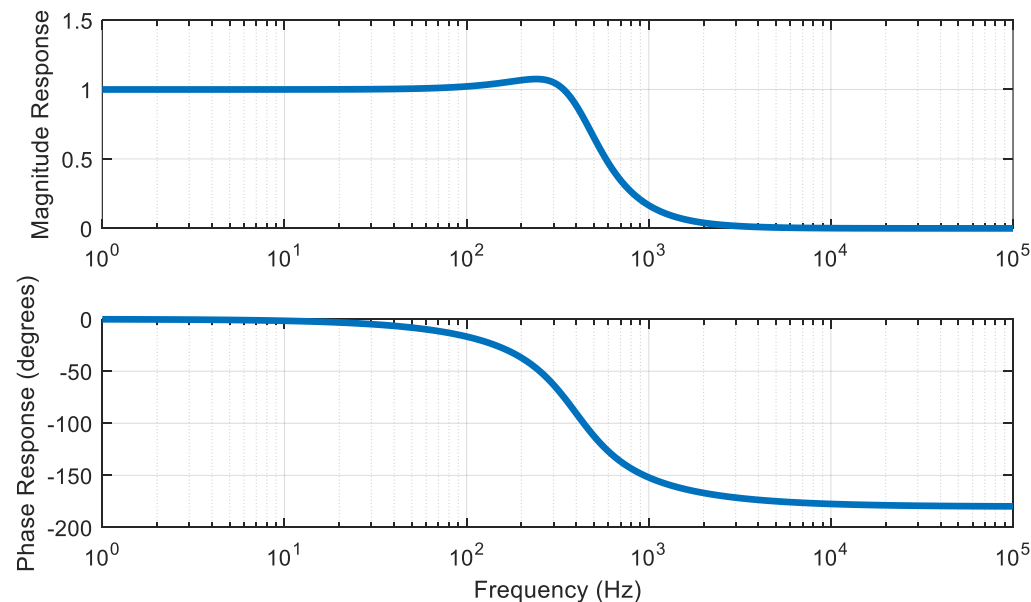
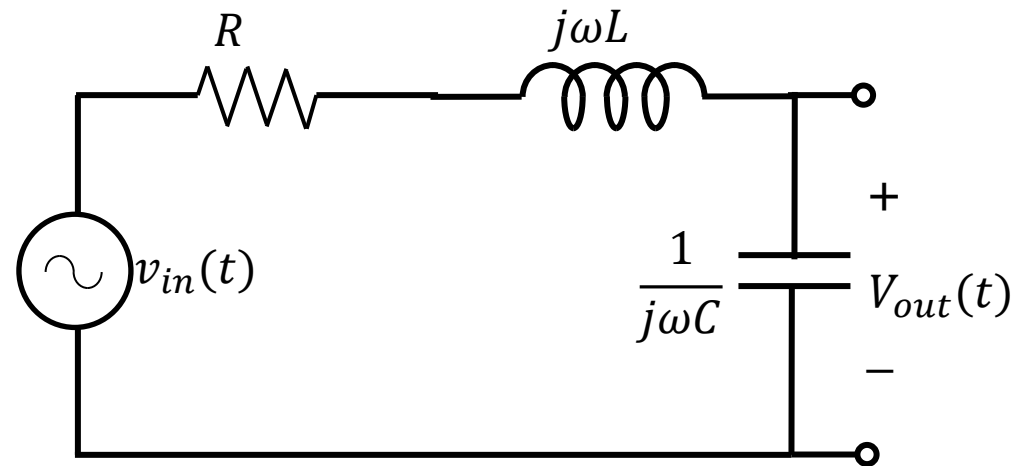
$$V_o = V_{in} \frac{\frac{1}{j\omega C}}{\frac{1}{j\omega C} + R + j\omega L}$$
$$= V_{in} \frac{1}{1 + j\omega RC - \omega^2 LC}$$

$$H(\omega) = \frac{V_o}{V_{in}} = \frac{1}{1 - \omega^2 LC + j\omega RC}$$

$$|H(\omega)| = \frac{1}{\sqrt{(1 - \omega^2 LC)^2 + (\omega RC)^2}}$$

$$\angle H(\omega) = -\tan^{-1} \left(\frac{\omega RC}{1 - \omega^2 LC} \right)$$

RLC Lowpass Filter



Application: Circuits



Example 8.5: Alternatively, for the series RLC, if the output voltage is measured on the resistor,

$$V_o = V_{in} \frac{R}{\frac{1}{j\omega C} + R + j\omega L}$$
$$= V_{in} \frac{j\omega RC}{1 + j\omega RC - \omega^2 LC}$$

$$H(\omega) = \frac{V_o}{V_{in}} = \frac{j\omega RC}{1 - \omega^2 LC + j\omega RC}$$

$$|H(\omega)| = \frac{\omega RC}{\sqrt{(1 - \omega^2 LC)^2 + (\omega RC)^2}}$$

$$\angle H(\omega) = \frac{\pi}{2} - \tan^{-1} \left(\frac{\omega RC}{1 - \omega^2 LC} \right)$$

RLC Bandpass Filter

